Analysis of Personal Datasets from Google, Spotify, Netflix and Swiggy

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***Abstract*— The aim of this project was to analyze my personal datasets from platforms such as Chrome, Spotify, Swiggy, and Netflix to identify lifestyle patterns and trends. The analysis shows differences between weekdays and weekends using **Netflix and Chrome** with **weekdays** there were about **9151** linked to productivity and **weekends** there were about **6157** more focused on entertainment. Morning and night routines using **Swiggy** and **Netflix** showed higher food orders of around **47** at night, while mornings were more of watching movies about **89**. Device usage using **Spotify, Netflix, Chrome** about **2270** Spotify reflects convenience and purpose, with mobile and Netflix about **752** used mostly, TV and tablet for entertainment, and Chrome search was around **8744** PC for education and career tasks. Year-level patterns highlighted life transitions, where **Chrome** and **Swiggy** peaked highest to **1799** and **16** respectively so during exam preparation and college work, while **Netflix** and **Spotify** peaked highest to **284** and **683** respectively so during holidays, showing leisure and relaxation.**

Keywords— Lifestyle patterns, Chrome, Spotify, Swiggy, Netflix

INTRODUCTION

This project focuses on analysis of 1my own dataset and analyzing my interaction with the apps first I chose around 6 datasets like chrome, apple, snapchat, ChatGPT, Netflix, Spotify but since few apps data was smaller, so I finally chose **Chrome, Swiggy, Spotify, Netflix** as I mostly used these apps recently. My main goal was to analyze my trends like weekday weekend interaction, Morning vs Evening pattern, device usage pattern and whole year interaction. For this I have used excel formula and plugins. This project is to analyze how I use each app for different purposes like food, Education, career, food habits and entertainment. It also showed the importance of keeping my data private while learning from it.

I. DATA

For this project I have initially considered the following 6 companies whose websites/apps I use often: Google, Apple, Spotify, Swiggy, Snapchat, Netflix. Out of them 4 personal datasets from different companies were selected and Google [1], Spotify [2], Netflix [3], Swiggy [4]

Table I presents a summary of the 4 personal datasets from 4 different companies that were selected for this project. The datasets were selected based on the following reasons:

- **Google(chrome):** This data was downloaded following the instructions from [1]. The dataset contains many apps and services, some of which I use very often. It provides information like site visited, time so I can analyze my search patterns and genre. However, after downloading and exploring the dataset I noticed that data was only available for 72 number of services, while for some other services the data was very small.
- **Spotify:** This data was downloaded following the instruction from [2]. This dataset also contained 1 service, but this was also one of my most used applications. So, this will give me useful and interesting data like records of tracks, artists, and timestamps, useful for analyzing my daily listening habits.
- **Netflix:** This data was downloaded following the instruction from [3]. However, the dataset contains only 1 service but which I use very often. It provides detailed information about my viewing patterns like entertainment consumption patterns, useful for analyzing time-of-day viewing habits.
- **Swiggy:** This data was downloaded following the instruction from [4]. This dataset contains 4 services which I used mostly when I was in India for Food ordering so this dataset will provide information like Timestamps, amount spent and food. So, this will help me to analyze my lifestyle pattern and food habits, but other 3 services data was small.

TABLE I. SUMMARY OF COMPLETE DATASETS FROM 4 COMPANIES.

Dataset Characteristic	Google	Spotify	Netflix	Swiggy
Number Apps/Services with Data	72	1	1	4
Example Apps/Services	Gemini, YouTube, Maps, Chrome, Calendar	Spotify music	Netflix entertainment	Food delivery, Instamart groceries, Dineout reservations, Genie pick-up & drop
Data Type	Structured, Semi-structured	Structured,Semi-structured	Structured	Structured
Dataset Size (Uncompressed)	18.1MB	2.56MB	498KB	2MB
Number Folders	32	1	13	4
Total Number Files	90	13	27	4
Number Files for Specific Formats	12(html), 53(json), 18(csv)	3(csv), 15(json)	1(json), 13(html)	4(csv)
Examples Data Files	bookmarks.html, My Activity.html, conference_history_retention.CSV	Marquee.json, stream_history.json,payments.json	Clickstream.csv, viewing_history.csv, viewing_activity.csv	Swiggy-order.csv, swiggy-ginie.csv, swiggy-instamart.csv
Timespan	2 years	1 year	2 years	1 year

TABLE II. SUMMARY OF SELECTED FILES.

Criteria	Google	Spotify	Netflix	Swiggy
Service Name	Chrome	Spotify	Netflix	Swiggy food delivery
File Name	My Activity.html	StreamingHistory_music_0.json	Viewing activity.csv	swiggy-orders-2025-11-28.csv
Size	7.89MB	456KB	262KB	10.4KB
File Format	HTML	JSON	CSV	CSV
Type	Semi-structured	Semi-structured	Structured	Structured
Number Files	1 HTML and 1 JSON	2 JSON	1 CSV	1 CSV
Num Attributes	5	4	10	6
Num Instances	11,307	3,095	1,419	89
Attributes List	Date,Time,visited,URL	endTime, artistName, trackName, msPlayed	Device,Title, Duration, Start Time, Country, Bookmark, Latest Bookmark	Amount, date , order no., place, item
Finding	Device pattern,weekday/weekend pattern and Monthly analysis	Monthly analysis and devices	Device pattern,weekday/weekend pattern, Morning and night pattern analysis and Monthly analysis	Morning and night pattern analysis and Monthly analysis
Correlation	Relation with Netflix, Spotify, Swiggy	Relation with Netflix, Google, Swiggy	Relation with Google, Spotify, Swiggy	Relation with Netflix, Spotify, Google

The challenges faced when downloading and exploring them include the following:

- **Google(chrome):** The process of downloading the data was easy but the main challenge I faced during exploring as the data was in HTML it was semi-structured, so I could only visually analyze the data and the no.of instances was around 11,000 so it was hard to analyze because of this I tried to convert the HTML file into CSV using excel data convert feature didn't work, then tried using Notepad++ plugin but it gave me all the instances in one single line, didn't work as well, then tried entering the value manually since it was around 11,000 instances it was time consuming at last I used JavaScript to convert it into CSV at first it gave me only the heading but when I added th tags it gave me CSV file. Second challenge was that the data didn't contain device information, so I needed to choose another file which had device information and it was in JSON, I converted it into CSV using Notepad++ plugin.
- **Spotify:** The downloading process of the data was simple, but it took 2 days to complete the request, the file was in JSON format so it was difficult to do analysis using graphs and table so I converted it into CSV using Notepad++ plugin which was not available so I downloaded, it converted it into text and have to copy and paste it in excel sheet then it was in CSV. This also didn't contain any device information which was necessary for my analysis, so I requested then converted into CSV from JSON.

  I had a idea of finding the genre of Spotify but it didn't come along with dataset so I was doing search how to add the genre of music and I thought of adding it by finding the song value and finding the mood of the song so I could relate it to Swiggy order no. for finding stress pattern so for the value of the song I need to use Spotify API website and get the credential apply it to postman API website by creating a workspace and i got the value for one code but I need to do it manually for each song but it was very hard to add for 4000 songs so it was time consuming and thought of other finding
- **Netflix:** The process of downloading was easy, and it was in CSV format with device information, but it had some unwanted attributes, so I needed to remove it.
- **Swiggy:** The data was available in the website couldn't be downloaded without a google extension, so I downloaded the google extension and downloaded the dataset

II. METHODOLOGY

The KDD methodology [6] was selected for this project as it has more detailed steps suitable for a project where the end result is written and presenting the results in a report.

1. Selection

Table II presents a summary of the data files selected from the 3 personal datasets.

2. Preprocessing

To convert **Chrome HTML** files to **CSV** first I needed to do some preprocessing like removing comma, &amp, [&nsbp] then th tags to convert into CSV. Then **Netflix** and **Swiggy CSV** data had date in "###" so I needed to make it into dd-mm-yyyy format also removed unwanted attribute like order no. For **Spotify JSON** file I downloaded JSON to CSV plugin in Notepad++

3. Transformation

For converting **Chrome** HTML to CSV, I used JavaScript in Notepad++. **Spotify** was in JSON then converted it into CSV using notepad++ plugin>JSON viewer tree> query CSV>then pasting in excel. **Swiggy** and **Netflix** were already in CSV so didn't transform it.

4. Data Mining

I saved all the CSV files to Excel to apply Excel formula for analysis. **Chrome** and **Netflix** used excel formulas to convert raw date into days using excel formula [5] then for all the dataset I applied UNIQUE formula to extract the instances, also applied COUNTIF for extracting the no.of occurrence of each instance. **Swiggy and Netflix** to extract Time of the day I removed the millisecond and applied formula [5].

5. Evaluation / Interpretation

I combined **Chrome** and **Netflix** made a table using excel formula to create weekday weekend pattern graph using pie chart. Then I correlated **Chrome**, **Netflix** and **Spotify** and made a table for analyzing the device pattern using bar graph. Also interrelated between **Swiggy** and **Netflix** to find Morning – Night pattern using pie chart. Finally, I related all the four Dataset to find the yearly analysis using bar graph.

III. Ethical Considerations

Ethical Approval: I got approval from my mother for downloading the Swiggy dataset as I couldn't have my own number since I was below 18, due to this I couldn't create an account. So, for this project I got consent from my mother, and I took this dataset because I was the primary user of that account. **Ethical consideration:** These datasets include personal details like what I watch, listen to, order, and browse. Because this information can reveal my habits and routines, there are privacy and ethical concerns if shared directly. To reduce risks, I removed names, addresses, full URLs, and exact times, and only used summaries such as monthly count. I also made sure the data only came from my own accounts and did not include information about other people. I also used my own laptop, and my Netflix data included my mail id, so I anonymized my mail id by removing the column.

IV. Findings

1)Analyzing Weekend and Weekday pattern:

To analyze my routine during my **Weekday and Weekend** I chose **Netflix and chrome** data as I was very active in both the platform and took day and no.of days then categorized into weekday and weekend.

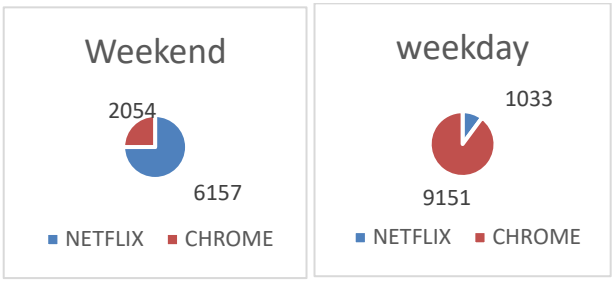


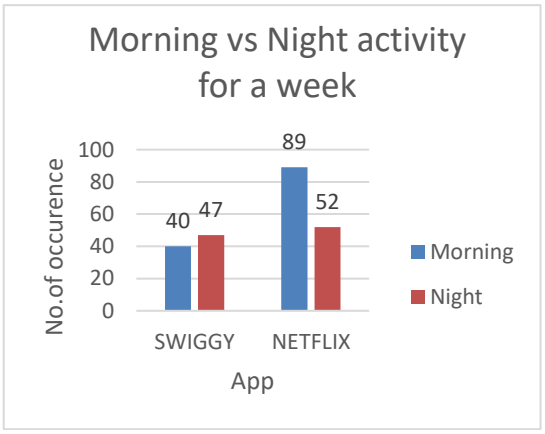
fig.1(Weekday vs Weekend)

From fig.1 I got to know that I spent Majority of my time on **Netflix** during **weekend** as I watched about **6157** movies and series over a year during weekends and I searched for about **9151** times in **chrome** during **weekdays** mostly it was about Moodle, NCI, past year paper, GitHub and career website. So, my weekend routine was mostly watching Netflix and enjoying, and weekday routine was about career and educational research.

2)Analyzing Morning vs Night pattern:

To analyze **Time of day** pattern I choose the platform that I am active during the whole day which was **Swiggy and Netflix** I took **Swiggy** data for each week and got the average as the **Netflix** value was higher it wouldn't be ideal to compare, and I did the same for Netflix and made a table out of it, I did this because I needed to find that did I do binge watching with food from restaurant.

fig.2(Morning vs Night activity)

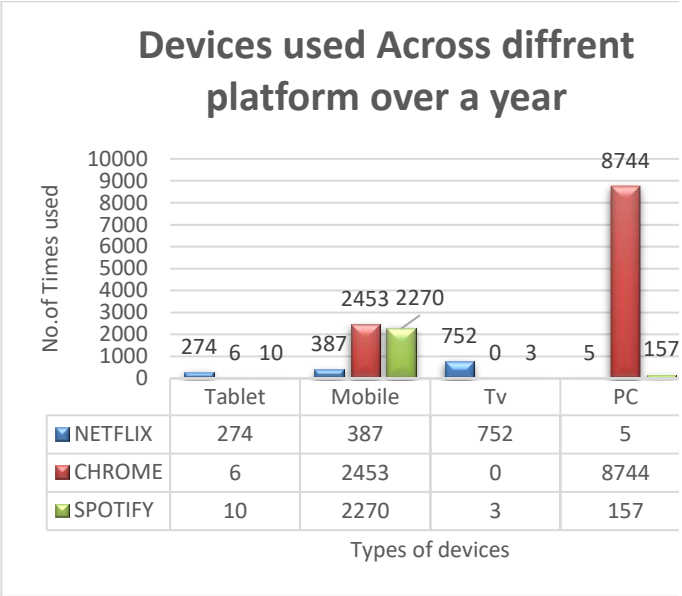


So, from the above graph I found out that night eating habits coincide with **Netflix** binge watching but I watched more movies during the daytime about **89** and ordered food mostly at night **52**. I watched comparatively less movies during Night and, but my Swiggy order pattern was much similar either ordered lunch or late-night dinner.

3)Analyzing Devices usage pattern:

To analyze most used devices on different platforms like **Netflix, Chrome and Spotify**. I took list of all the registered devices and categorized them into **Mobile, TV, PC, Tablet**.

fig.3(device usage)

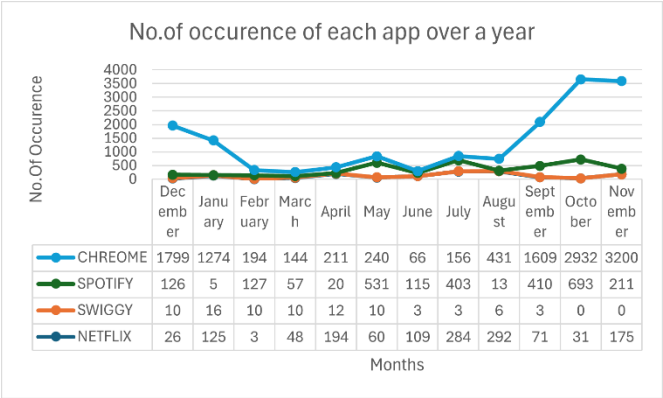


From the above graph I have listened to **4270** songs on **Spotify** high when compared to other devices on the **Mobile**, so this was because I listened to songs mostly when I traveled so it was handy and easy to use, whereas I have watched about **752** movies on **Netflix** when compared to other devices **Tv** was high, this shows that watching movies on big screen for comfortability, on other hand I have searched about **8744** on **Chrome** when compared to other devices **PC** was high, this shows that I have searched and all work related was on PC. All apps used other apps too, but these devices dominated.

4)Analyzing Year-level pattern:

To Analyze my whole year pattern, I took all the 4 datasets and saw their usage for 12 Months I used **Swiggy, Chrome, Netflix, Spotify**.

fig.4(year- level pattern)



From the above graph I have used **Chrome** highest activity overall, with **1799 in December** and **1274 in January**, but then it reduces in the upcoming month as I had my exams during December and January then **Spotify** rise, from **126** in December and **5** in January, but high in May with **531** and again in July with **403**, **Netflix** usage was unstable, with **194** in April and another rise in July with **284** , as I listened and watched lot songs and movies during summer as I had my holidays. **Swiggy** stayed low throughout, with small increases like **16** in January highest of all months and **12** in March as I had my exam and I was stress eating. This shows that I used **Chrome** mostly for education purpose as it peaked during my exam January and when I was preparing for my college was linked more to productivity, while I used Spotify and Netflix highly during relaxed periods.

V. Conclusions and Future Work

The aim of this project was to analyze my personal data and find my lifestyle pattern, and their trend fig.1 shows my weekend weekday pattern weekday is more productive whereas weekend is more into entertainment, fig.2 shows my Morning vs Night pattern where I ordered and ate lot of food during night and watched movies and I also watched movies in the Morning. fig.3 this shows that I have used mobile for my convenience and tablet and Tv for watching and entertainment, PC was for education and career. ig.4 shows my life transition as chrome and Swiggy peaked when I had my exam and when I was preparing for my college whereas Netflix and Spotify peaked when I was on my holiday, If I had more time I would have built a machine learning project and prediction.

References

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